

Refining of Petroleum :-

The process of separation of crude oil into different useful fractions on the basis of their boiling points is called petroleum refining or fractionation.

The crude oil is in the form of oil-water emulsion. Emulsion is broken by either preheating to 50°C or by adding demulsifying agents.

The emulsion free crude oil is heated to $350^{\circ}\text{--}400^{\circ}\text{C}$, at which crude oil is almost converted into vapours. The vapours are then fed into a specially designed bubble cap fractionating column.

The different hydrocarbons boil at different temperatures according to their size and molecular weight. The superheated steam is simultaneously fed into the column. The fractionating column permits separation of petroleum vapours into a number of fractions having definite boiling point range.

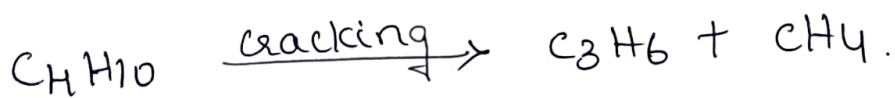
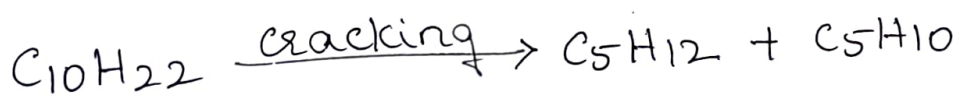
The more volatile component condenses in the upper plates of the fractionating column while the less volatile fraction is collected at the lower plates. The various fractions condensed at different parts of the columns are continuously withdrawn and some of them are further fractionated and processed.

Cracking - (Chemical processing of petroleum) ⑦

Primary distillation (Refining) of crude oil yields only about 20% of petrol. The major fractions obtained are heavier fractions. These heavier fractions are broken down to smaller molecules to produce more useful lighter fractions. The process is known as cracking of petroleum.

"Cracking is a process in which the long chain hydrocarbon molecules are split into smaller fragments."

During cracking process the C-C and C-H bonds in the chain are ruptured yielding the molecules with low molecular weight.



The cracking is done by two types of processes
1) Thermal cracking 2) Catalytic cracking.

1) Thermal cracking -

In thermal cracking, the feed stock heavy oil is subjected to high pressure and temperature (upto 700°C) in the absence of air and catalyst.

2) Catalytic cracking

A major advantage of catalytic cracking over thermal process is that the resulting products contains a higher percentage of aromatic hydrocarbons like benzene and a lower percentage of alkenes. Cracking is usually carried out in the presence of catalyst at milder conditions i.e., low temperature and low pressure. Therefore there should be intimate contact between the reactant vapour and the solid catalyst. Therefore catalyst in the form of powder or pellets having large surface area is more useful.

The catalytic cracking is carried out by two methods -

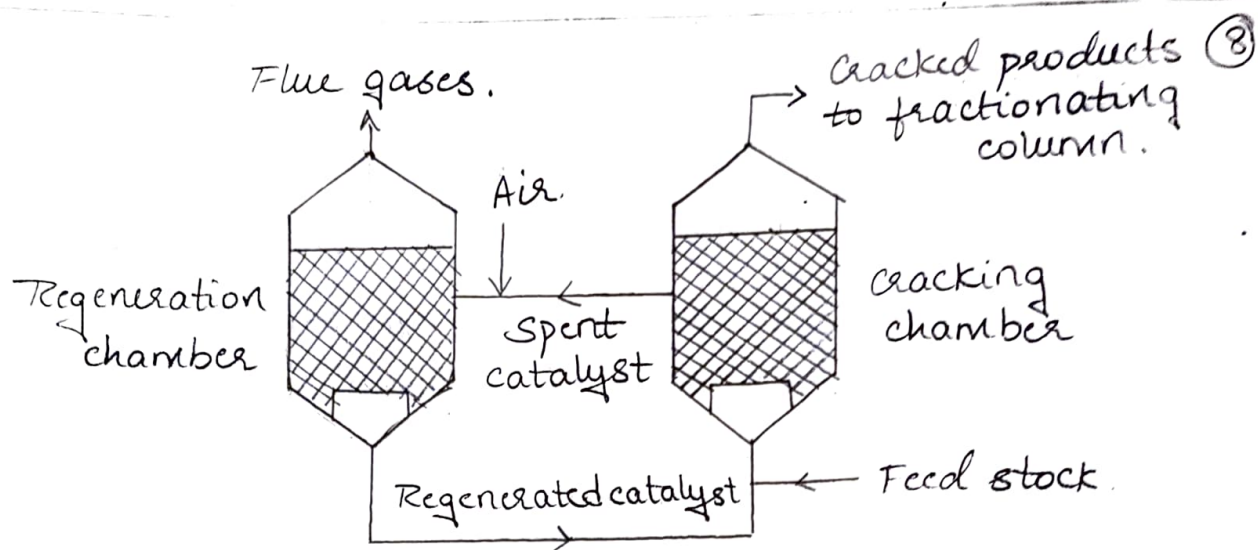
(i) Fixed Bed catalytic cracking process

In this cracking process, the production has to be interrupted in order to regenerate the inactive catalyst. Interrupting of the production line being expensive, continuous process have been developed in catalytic cracking where by the regeneration of catalyst is carried out continuously without interfering with gasoline production.

(ii) Moving (Fluidised) bed catalytic cracking

Process : optimum conditions

1. Feed stocks :- Gas oil, Heavy oil fractions
2. Catalyst :- $\text{Al}_2\text{O}_3 + \text{SiO}_2$ (fluidized)
3. Temperature :- 550°C
4. Pressure :- little above the normal pressure
5. Production yield :- 106 gallons per day.



This is the most modern and widely used method of catalytic cracking. Heavy oil pumped through preheater to a pipe is vapourised. The vapour of oil is mixed with finely powdered catalyst. The fluid so formed is passed into the cracking chamber at about 550°C . Cracked vapours are withdrawn from the top of the cracking chamber and directly fed into a fractionating column to separate into gases, gasoline, gas oil and uncracked oils. The uncracked oil may be cracked in a second stage cracking, thereby increasing the overall yield of the cracked products.

The particles of the catalyst becomes slightly heavy due to the deposition of carbon and comes out of the reactor. A stream of air blown over the hot catalyst, takes the catalyst into regeneration chamber in which elemental carbon deposited on catalyst surface is burnt off. The regenerated catalyst is mixed with fresh feed stock and returned to the cracking chamber.

Reforming of petrol :-

Gasoline used in automobiles mainly consists of a mixture of hydrocarbons - alkanes, cycloalkanes and aromatics. The octane number of hydrocarbons increases as the hydrocarbons become more compact. It is observed that branched chain hydrocarbons have higher octane numbers than the corresponding straight chain hydrocarbons. Aromatic hydrocarbons have higher octane number than non-aromatic hydrocarbons.

The gasoline obtained from primary distillation of crude oil contains mainly straight chain hydrocarbons. As such, it has a very low octane rating, not greater than 60. But internal combustion engines require gasoline of octane number above 80. Therefore to increase the octane number of hydrocarbons i.e., to upgrade the octane number of petrol fraction, Reforming process is important.

The process involves a molecular rearrangement of hydrocarbons without any change in the number of carbon atoms to form new compounds.

Reforming is carried through structural modifications such as conversion of straight chain hydrocarbons into branched, cyclic and aromatics. This process is called catalytic reforming as it is carried out in presence of the catalyst.

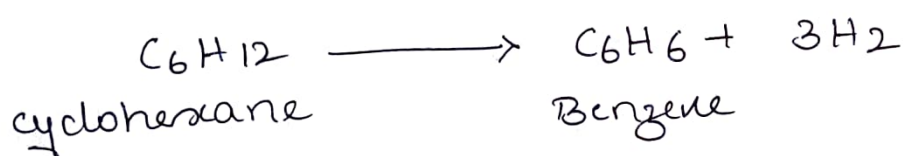
The main reactions taking place during catalytic reforming process are - isomerisation, Dehydrogenation, cyclisation etc.

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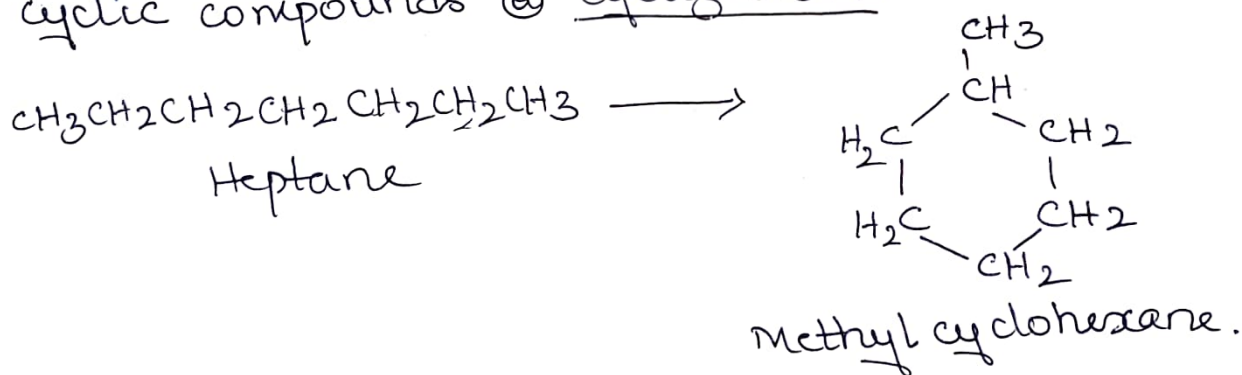
- 1) conversion of straight chain hydrocarbons into branched chain hydrocarbons @ Isomerisation.



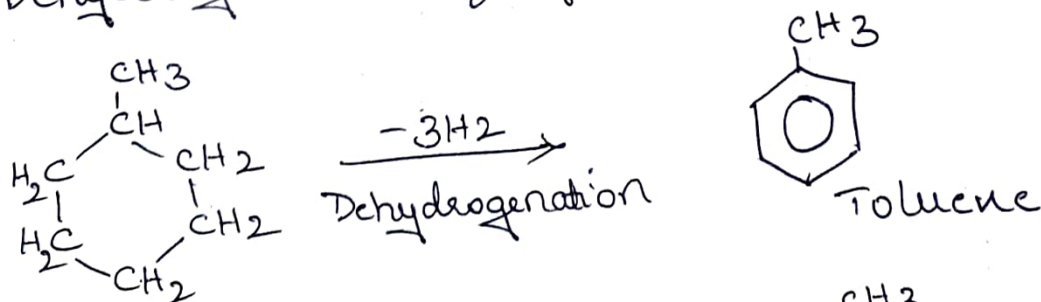
- 2) conversion of cycloalkanes into aromatic compounds @ Dehydrogenation.



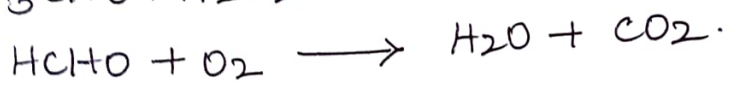
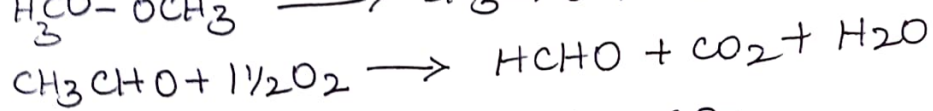
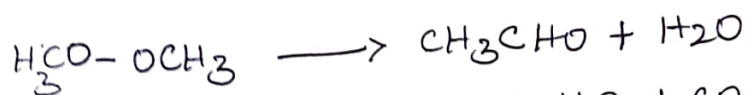
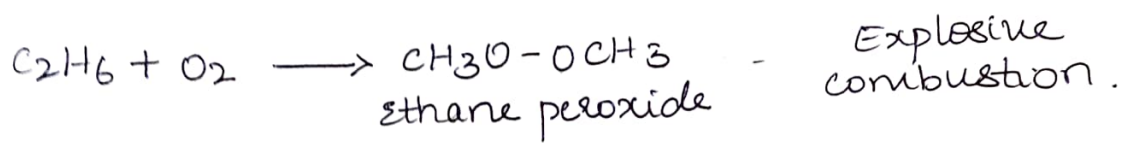
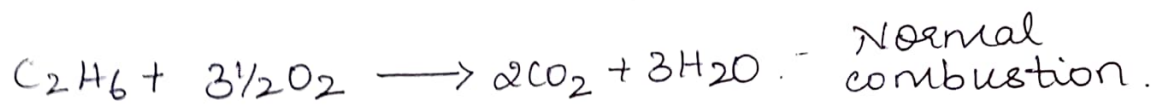
- 3) conversion of straight chain hydrocarbons into cyclic compounds @ cyclization.



- 4) Dehydrogenation of cycloalkanes.



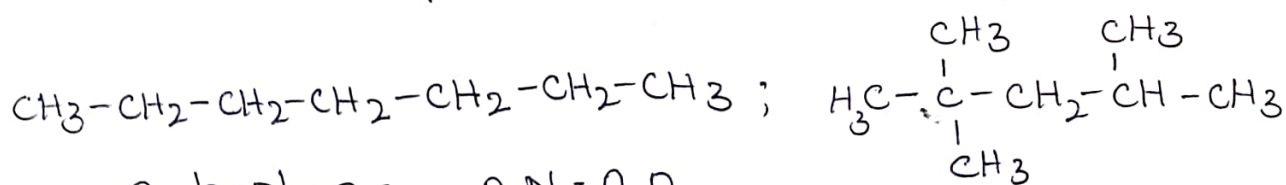
- 5) $2\text{CH}_3-\underset{\text{CH}_2}{\underset{\text{||}}{\text{C}}}-\text{CH}_3 \xrightarrow{\text{Hydrogenation}} \text{CH}_3-\underset{\text{CH}_3}{\underset{|}{\text{C}}}-\text{CH}_2-\underset{\text{CH}_3}{\underset{|}{\text{CH}}}-\text{CH}_3$
- isobutane isooctane.



Octane number 1-

Octane numbers of gasolines are simply defined with reference to n-heptane (C_7H_{16}) and isooctane (C_8H_{18}), the two standards chosen.

The straight chain hydrocarbon, n-heptane, which has poor combustion characteristics and knocks badly is arbitrarily given an octane number of 'zero', whereas the branched chain hydrocarbon, isooctane (2,2,4-trimethylpentane), which has an excellent combustion characteristics and very little tendency to knocking is given an octane number of 100.



n-heptane, O.N = 0.0

isooctane, O.N = 100

The octane number of a gasoline is "the percentage volume of isooctane in a mixture of isooctane and n-heptane blend, which has the same knocking characteristic as the gasoline sample under test". The different gasoline samples are rated by their octane numbers. Higher the octane number of a gasoline, least is its knocking tendency and higher is its combustion characteristics.

It has been found that knocking tendency is 11 largely related to chemical structures of fuels. The decreasing tendency of fuels to knocking is as follows-

straight chain hydrocarbons > Branched chain hydrocarbons > cycloalkanes > Alkenes > Aromatics.

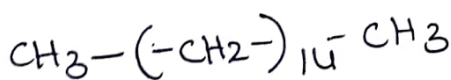
2, 3-Dimethyl pentane has been assigned an octane number 75, because it has the knocking tendency as that of a mixture of n-heptane and iso-octane containing 75% by volume of isooctane.

Cetane number:-

The knocking characteristics of a diesel oil is expressed in terms of cetane number. The suitability of a diesel fuel is determined by its cetane value.

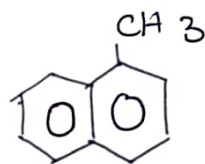
straight chain compounds undergo easy compression ignition and n-cetane is chosen as the upper limit of cetane number - 100.

Branched chain and cyclic compounds do not undergo compression ignition readily and 1-methyl naphthalene is chosen as the lower limit of cetane number = 0.



n-cetane

CN = 100



1-methyl naphthalene
CN = 0.

Cetane number of a diesel is "the percentage by volume of n-cetane in a mixture of n-cetane and 1-methyl naphthalene that gives the same performance as that of diesel fuel under test".